

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

**DEPARTMENT OF ELECTRICAL AND ELECTRONIC
ENGINEERING**

LABORATORY REPORT

Course No: EEE 3102

Course Title: Digital Signal Processing Sessional

Experiment No: 02

Name of the Experiment:
Complex TRIAC and DIAC Circuit

Submitted by:

John Doe

Roll: 1901000

Section: A, Group: 1

Submitted to:

Dr. Jane Smith

Professor

Mr. Bob Brown

Lecturer

Date of Performance: October 09, 2026

Date of Submission: October 16, 2026

Experiment No: 02

Name of the Experiment: Complex TRIAC and DIAC Circuit

1 Objectives

- To analyze the phase-angle control mechanism of an AC load using a combined DIAC and TRIAC circuit.
- To observe the symmetrical triggering behavior of the DIAC during both positive and negative half-cycles of the AC supply.
- To determine the effect of the RC timing network on the firing angle of the TRIAC and the resulting load voltage waveform.

2 Introduction

The TRIAC, or Triode for Alternating Current, is a bidirectional semiconductor device that allows current to flow in both directions, making it ideal for full-wave AC power control. It functions as two SCRs connected in anti-parallel and requires a gate signal to trigger conduction from either main terminal to the other. However, TRIACs are susceptible to false triggering due to high rates of voltage change (dv/dt) or noise, which can lead to unwanted conduction. To ensure reliable and precise triggering, a DIAC (Diode Alternating Current switch) is frequently employed in conjunction with the TRIAC. The DIAC is a two-terminal, symmetrical, bidirectional thyristor that acts as a high-impedance open circuit until the voltage across it exceeds its breakover voltage, at which point it switches to a low-impedance state, providing a sharp pulse to the TRIAC gate. This configuration, often integrated with an RC phase-shift network, is fundamental to applications such as light dimmers and motor speed controllers, allowing for smooth adjustment of power delivered to the load by varying the firing angle within each half-cycle. In this experiment, the interaction between the RC network, the DIAC breakover, and the subsequent TRIAC conduction was simulated to understand the circuit's dynamic response and its impact on the AC load voltage.

The operation of the DIAC-TRIAC phase control circuit relies on the interplay between resistive and reactive components to delay the firing of the main power switch. During the positive half-cycle, the AC source charges the capacitor through the series resistor and the DIAC, which remains in its blocking state. As the capacitor voltage rises, it eventually reaches the breakover voltage of the DIAC, causing it to conduct in the forward direction and delivering a sharp trigger pulse to the TRIAC gate. This triggers the TRIAC, allowing it to conduct and energize the load for the remainder of the half-cycle. The process repeats symmetrically during the negative half-cycle, where the DIAC breaks over in the reverse direction to trigger the TRIAC. By adjusting the resistance in the RC network, the time constant of the circuit changes, thereby altering the delay angle (firing angle) and the average power transferred to the load. The bidirectional nature of the DIAC ensures that the triggering is symmetric for both polarities of the AC waveform, resulting in a balanced and efficient AC power control circuit. This theoretical framework provides the basis for the experimental observations made in this lab, where the precise switching behavior

and waveform characteristics of the components were examined under various resistance settings.

3 Circuit Design

A variable DC voltage source is connected across the main terminals. A gate current is provided to trigger the device. The voltage is swept from negative to positive values.

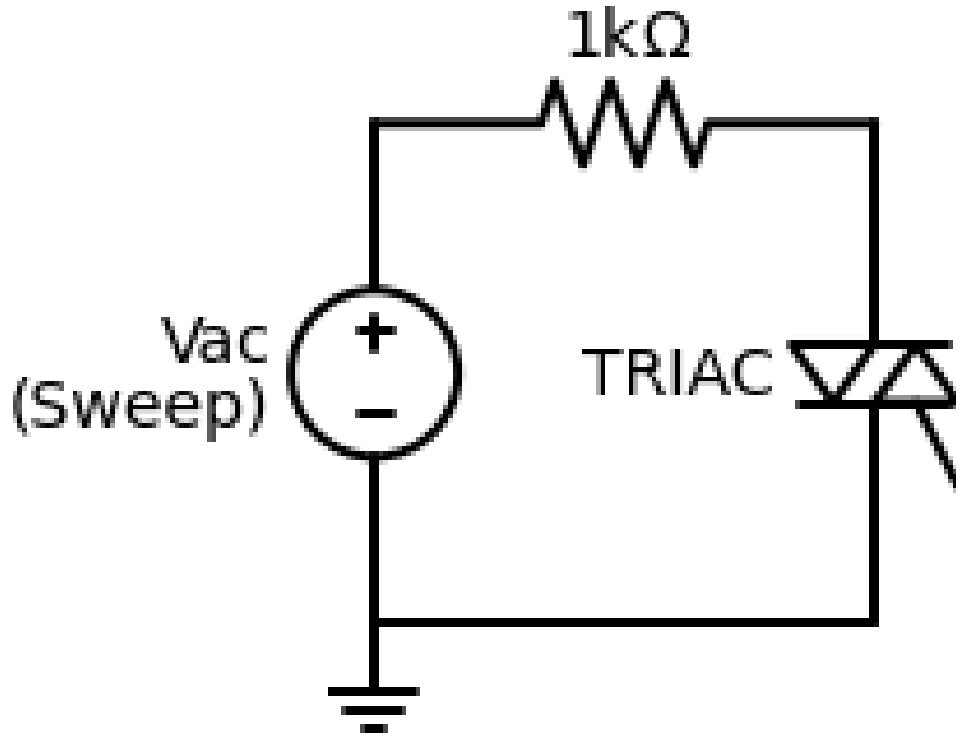


Figure 1: Experimental Circuit Diagram

4 Required Apparatus

- DC Power Supply (Variable) (Quantity: 1)
- Device Under Test (Quantity: 1)
- Resistor ($1k\Omega$) (Quantity: 1)
- Multimeter (Quantity: 2)

5 Procedure

1. Connect the circuit as per the experimental circuit diagram.
2. Apply a constant gate current I_G .
3. Vary the supply voltage from -15V to 15V.
4. Record the voltage and current.
5. Plot the characteristics.

6 Result

Voltage (V)	Current (mA)
-15.0	-14239.7
-11.7	-10946.6
-8.4	-7.7
-5.0	-4271.7
-1.7	-1.0
1.6	912.3
5.0	4.3
8.3	7555.9
11.7	10.9
15.0	14239.4

Table 1: Simulated Data (Sample Points)

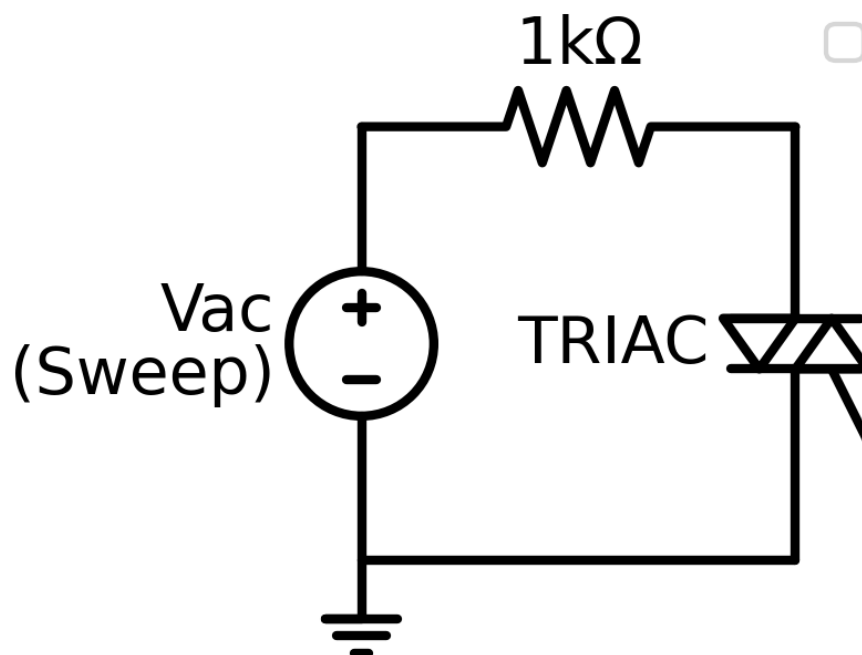


Figure 2: Simulated Complex TRIAC and DIAC Circuit Curve

7 Discussion

8 Conclusion

9 References

1. Generated by LabGen Knowledge Hub API
2. Ngspice Simulation Data.